

Heaven's Light of Our Guide
Rajshahi University of Engineering & Technology
Department of Mechatronics Engineering
Assignment (MTE 2020 Series)

Course No.: MTE 4101

Mark: 10

Course Title: Automation

Deadline: Before the 4th Year Odd Semester Board Viva

CO- Course Outcome (CO1), **PO-** Program Outcome (PO1)

Title: Advanced Automated Bottling Plant with Integrated Quality Control, Parallel Processing & Fault Recovery.

Objectives: This PLC-controlled plant should handle the entire cycle from bottle infeed to packaging, with multiple conveyors, parallel operations, automatic fault handling, and quality inspection.

Stage 1: Bottle Feeding & Detection

- Bottles arrive from storage conveyor. Entry gate opens only when the filling station is ready.
- Photoelectric sensor detects missing bottles. If gap detected, skip filling sequence for that position.

Stage 2: Washing Stage

- Bottles pass through automatic washer. Rotary spray arms activated only when bottle detected inside chamber.
- Timer ensures proper wash duration. Drain pump removes used water.

Stage 3: Filling Stage

- When the bottles reach to the filling station, a filling pump will active for filling the bottles.
- The station uses level sensors for precise filling.

Stage 4: Capping Stage

- Automatic capping head places caps.
- Cap presence sensor ensures cap is applied; if missing, bottle marked for rejection.

Stage 5: Labeling Stage

- Label applicator triggered only if bottle passes both fill & cap quality checks.

Stage 6: Packaging & Palletizing

- Bottles grouped in sets of 12.
- Two robotic arms (Consider two motor) operate in parallel- one loads bottles into boxes, the other palletizes boxes.

Stage 7: Fault Detection & Recovery

- If any station stops (jam, low pressure, empty cap feeder, low label roll), PLC pauses upstream processes while allowing downstream processes to finish.

Tasks:

1. Describe about automation, its importance in Industry 4.0, State of automation, challenges to implement automation and possible solutions to solve these problems in case of Bangladesh,
2. Draw I/O mapping table.
3. Create process flowchart for entire system.
4. Write step-by-step sequential logic for each stage.
5. Implement ladder diagram in PLC simulation software.
6. Demonstrate system in simulation with:
 - Normal operation.
 - Missing bottle scenario.
 - Overflow scenario.
 - Fault scenario (jam, low pressure, cap feeder empty).
7. Address the societal and environmental impact of automation.
8. Conclusion

Assignment Rubrics

Criteria	Excellent Mark: 02	Proficient Mark: 1.5	Good Mark: 01	Need to Improve Mark: 0.50
Understanding of the Problem	1. Demonstrates an in-depth understanding of automation and applies them effectively to solve complex engineering problems.	1. Demonstrates good understanding with minor inaccuracies in application.	1. Demonstrates basic understanding with significant gaps in application.	1. Lacks understanding of concepts or misapplies principles entirely.
Problem Analysis	1. Accurately identifies complex engineering problems, 2. Formulates the problem statement clearly. 3. Proposes appropriate methods for solving.	1. Accurately identifies complex engineering problems, 2. Formulates the problem statement with minor clarity 3. Proposes less appropriate methods for solving.	1. Problem identification is vague, 2. Proposed methods are incomplete and unclear.	1. Fails to identify problems or proposes inappropriate methods.
Circuit Design and Implementation	1. Circuit/system design is logical, and meets requirements, 2. Design is fully functional, backed by engineering knowledge.	1. Circuit/system design meets most requirements, with minor functional or design flaws.	1. Circuit/system design is partially functional or incomplete.	1. Design is non-functional, does not meet requirements, or lacks effort.
Result and Performance Analysis	1. All result, and calculations are accurate, 2. Results are thoroughly explained, and effectively support the solution.	1. All result, and calculations are mostly accurate, 2. Results, calculations are mostly accurate with minor errors or lack of clarity in explanation.	1. Analysis is incomplete, frequently incorrect, or poorly explained.	2. Analysis is missing, largely incorrect, or irrelevant.
Documentation, Report, and Timeliness	1. Comprehensive, well-organized report with detailed explanations, diagrams, and references. 2. Submitted on time, follows guidelines, and is professionally presented.	1. Complete report with adequate explanations and diagrams, though minor improvements needed in clarity. 2. Submitted on time with minor deviations in formatting or guidelines.	1. Report is incomplete, poorly organized, or missing key elements. 2. Submission is late or partially follows guidelines.	1. Report is missing or lacks meaningful content. 2. Submission is late and poorly formatted or presented.